

1. **(150 points)** Design and create a data warehouse for the database provided.

This data warehouse is designed to track and analyze film performance across the company's rental operations. Each row in the warehouse represents a single film and consolidates information drawn from multiple transactional tables into one analytical dataset. The warehouse integrates data from film, inventory, rental, payment, customer, category, and film actor tables for a complete view of how each film performs in terms of demand, revenue generation, inventory distribution, and customer engagement. By aggregating this information, the warehouse enables business users to evaluate which films generate the most revenue, which titles are rented most frequently, and how efficiently inventory is utilized across stores.

```
CREATE TABLE dw_film_performance AS
SELECT
    f.film_id,
    f.title,
    f.release_year,
    f.rating,
    f.length,
    f.rental_rate,
    c.name AS category_name,

    COUNT(DISTINCT cust.customer_id) AS customer_count,
    ROUND(COUNT(DISTINCT CASE WHEN cust.active = 1 THEN cust.customer_id END)
        / COUNT(DISTINCT cust.customer_id), 2) AS active_cust_rate,

    COUNT(DISTINCT fa.actor_id) AS actor_count,

    COUNT(DISTINCT i.inventory_id) AS inventory_count,
    COUNT(DISTINCT i.store_id) AS stores_stocking_count,

    COUNT(DISTINCT r.rental_id) AS rental_count,
    SUM(p.amount) AS total_revenue,
    ROUND(AVG(p.amount),2) AS avg_payment,
    MIN(r.rental_date) AS first_rental_date,
    MAX(r.rental_date) AS last_rental_date,

    ROUND(AVG(CASE WHEN r.return_date IS NULL THEN NULL
        ELSE DATEDIFF(r.return_date, r.rental_date)
    END),2) AS avg_days_out,

    ROUND((COUNT(DISTINCT r.rental_id) / NULLIF(COUNT(DISTINCT i.inventory_id),0)),2) AS
    rentals_per_copy

FROM film f
    JOIN film_category fc ON fc.film_id = f.film_id
    JOIN category c ON c.category_id = fc.category_id

    LEFT JOIN film_actor fa ON fa.film_id = f.film_id
    LEFT JOIN inventory i ON i.film_id = f.film_id
    LEFT JOIN rental r ON r.inventory_id = i.inventory_id
    LEFT JOIN payment p ON p.rental_id = r.rental_id
    JOIN customer cust ON cust.customer_id = r.customer_id

GROUP BY
```

f.film_id, f.title, f.release_year, f.rating, f.length, f.rental_rate,
l.name, c.name;

film_id	title	release_year	rating	length	rental_rate	category_name	customer_count	active_cust_rate	actor_count	inventory_count	stores_stocking_count	rental_count	total_revenue	avg_payment	first_rental_date	last_rental_date	avg_days_out	rentals_per_copy
1	ACADEMY DINOSAUR	2006	PG	86	0.99	Documentary	23	0.96	10	7	2	23	367.70	1.60	2005-05-27 07:03:28	2005-08-23 01:01:01	5.09	3.29
2	ACE GOLDFINGER	2006	G	48	4.99	Horror	7	0.86	4	3	1	7	211.72	7.56	2005-07-07 19:46:51	2006-02-14 15:16:03	5.67	2.33
3	ADAPTATION HOLES	2006	NC-17	50	2.99	Documentary	12	1.00	5	4	1	12	109.40	3.16	2005-05-31 04:50:07	2005-08-23 13:54:39	3.42	3.00
4	AFFAIR PREJUDICE	2006	G	117	2.99	Horror	22	1.00	5	7	2	23	458.85	3.99	2005-05-27 20:44:36	2006-02-14 15:16:03	4.73	3.29
5	AFRICAN EGG	2006	G	130	2.99	Family	12	1.00	5	3	1	12	259.40	4.32	2005-05-28 07:53:38	2006-02-14 15:16:03	7.09	4.00
6	AGENT TRUMAN	2006	PG	169	2.99	Foreign	21	0.95	7	6	2	21	887.53	6.04	2005-05-26 15:32:46	2005-08-21 16:03:01	5.90	3.50
7	AIRPLANE SIERRA	2006	PG-13	62	4.99	Comedy	15	1.00	5	5	2	15	414.25	5.52	2005-06-20 21:11:50	2005-08-22 17:18:05	4.53	3.00
8	AIRPORT POLLOCK	2006	R	54	4.99	Horror	18	1.00	4	4	1	18	411.28	5.71	2005-05-25 19:37:47	2005-08-23 20:24:36	5.06	4.50
9	ALABAMA DEVIL	2006	PG-13	114	2.99	Horror	12	1.00	9	5	2	12	646.92	5.99	2005-07-06 18:32:49	2005-08-23 14:26:51	5.67	2.40
10	ALADDIN CALENDAR	2006	NC-17	63	4.99	Sports	23	1.00	8	7	2	23	1054.16	5.73	2005-06-15 11:03:24	2005-08-21 20:49:21	4.91	3.29
11	ALAMO VIDEOTAPE	2006	G	126	0.99	Foreign	23	0.96	4	7	2	24	143.04	1.49	2005-05-28 09:10:49	2005-08-22 04:04:31	5.08	3.43
12	ALASKA PHANTOM	2006	PG	136	0.99	Music	24	0.96	7	7	2	26	313.18	1.72	2005-05-29 21:58:43	2005-08-23 06:11:52	5.27	3.71
13	ALT FOREVER	2006	PG	150	4.99	Horror	9	1.00	5	4	1	9	274.55	6.10	2005-07-12 10:51:09	2006-02-14 15:16:03	4.63	2.25
15	ALIEN CENTER	2006	NC-17	46	2.99	Foreign	22	0.95	6	6	2	22	544.68	4.13	2005-05-27 22:49:27	2005-08-23 19:21:22	4.91	3.67
16	ALLEY EVOLUTION	2006	NC-17	180	2.99	Foreign	14	0.93	5	4	2	14	264.30	3.78	2005-05-25 09:21:29	2005-08-22 07:10:26	5.57	3.50
17	ALONE TRIP	2006	R	82	0.99	Music	18	1.00	8	6	2	18	502.56	3.49	2005-05-29 04:35:29	2006-02-14 15:16:03	5.47	3.00
18	ALTER VICTORY	2006	PG-13	57	0.99	Animation	22	0.91	4	6	2	22	131.12	1.49	2005-05-28 21:31:08	2005-08-23 03:44:51	4.68	3.67
19	AMADEUS HOLY	2006	PG	113	0.99	Action	21	1.00	6	6	2	21	202.74	1.61	2005-05-28 12:08:37	2006-02-14 15:16:03	4.85	3.50
20	AMELIE HELLFIGHTERS	2006	R	79	4.99	Music	10	1.00	6	3	1	10	407.40	6.79	2005-06-17 13:46:52	2006-02-14 15:16:03	4.80	3.33
21	AMERICAN CIRCUS	2006	R	129	4.99	Action	20	1.00	5	6	2	22	838.90	7.63	2005-05-25 21:58:58	2006-02-14 15:16:03	5.43	3.67
22	AMISTAD MIDSUMMER	2006	G	85	2.99	New	21	1.00	4	7	2	21	283.16	3.37	2005-06-15 07:29:59	2006-02-14 15:16:03	4.45	3.00
23	ANACONDA CONFES...	2006	R	92	0.99	Animation	21	0.95	5	5	2	21	303.95	2.89	2005-05-30 19:20:03	2005-08-20 21:21:08	4.38	4.20
24	ANALYZE HOOSIERS	2006	R	181	2.99	Horror	14	1.00	5	4	1	14	279.30	3.99	2005-06-16 13:59:42	2005-08-22 17:58:27	5.71	3.50
25	ANGELS LIFE	2006	G	74	2.99	New	21	0.95	9	6	2	22	1078.02	5.44	2005-05-27 16:21:26	2005-08-23 13:46:00	5.14	3.67

2. (100 points) Create *eight* SQL queries on your data warehouse (not on the original dataset)

Query # 1: Top Films by total revenue:

Identifies the company's highest-earning films to prioritize marketing, retention, and acquisition of similar titles.

The company should prioritize marketing and visibility for the top revenue- generating films and consider acquiring similar titles in those genres, as they demonstrate proven profitability and strong customer demand.

```
SELECT
    film_id, title, category_name,
    total_revenue, rental_count
FROM dw_film_performance
ORDER BY total_revenue DESC
LIMIT 25;
```

film_id	title	category_name	total_revenue	rental_count
892	TITANIC BOONDOCK	Animation	1857.24	23
764	SATURDAY LAMBS	Sports	1842.48	28
369	GOODFELLAS SALUTE	Sci-Fi	1677.52	31
879	TELEGRAPH VOYAGE	Music	1622.11	27
938	VELVET TERMINATOR	Comedy	1599.66	26
680	PINOCCHIO SIMON	Comedy	1471.80	20
893	TITANS JERK	Sci-Fi	1411.97	29
771	SCORPION APOLLO	Drama	1350.16	23
973	WIFE TURN	Documentary	1342.14	31
551	MAIDEN HOME	New	1310.08	24
263	DURHAM PANKY	Sports	1302.66	26
827	SPICE SORORITY	Drama	1273.80	20
320	FLAMINGOS CONNE...	New	1257.93	23
239	DOGMA FAMILY	Animation	1250.90	30
172	CONEHEADS SMOOCHY	Drama	1230.75	25
403	HARRY IDAHO	Drama	1174.20	30
414	HELLFIGHTERS SIERRA	Foreign	1173.60	20
603	MOVIE SHAKESPEARE	Family	1165.92	27
694	PREJUDICE OLEANDER	Classics	1158.00	20
112	CALENDAR GUNFIGHT	Foreign	1134.32	21
846	STING PERSONAL	New	1118.53	21
243	DOORS PRESIDENT	Animation	1114.29	19
48	BACKLASH UNDEFEA...	Children	1111.67	19
785	SHAWSHANK BUBBLE	Travel	1098.10	19
361	GLEAMING JAWBREA...	Sports	1093.60	29

Query #2: Most rented films and actor count:

This query explores which films are rented more frequently and whether films with more actors are more popular - does a star-studded cast draw a variety of fans to a movie? These results assist in

deciding which titles to prioritize when purchasing inventory, and whether they should plan to purchase more copies or set higher rental prices for newly released films that have more actors.

SELECT

film_id, title, category_name,
rental_count, total_revenue, actor_count

FROM dw_film_performance

ORDER BY rental_count DESC

LIMIT 25;

Query #3: Best revenue per rental (pricing/ premium performance)

Finds films generating unusually high revenue per rental, which can signal opportunities for premium pricing, customer willingness to rent movies for longer durations, revenue generated from late fees or other profit markers. Films with strong revenue per rental may be better candidates for extended promotional placement, and similar films could be priced strategically to maximize profitability.

SELECT

film_id, title, category_name,
rental_count, total_revenue,

ROUND((total_revenue /
NULLIF(rental_count,0)),2) AS
revenue_per_rental

FROM dw_film_performance

WHERE rental_count >= 20

ORDER BY revenue_per_rental DESC

LIMIT 25;

film_id	title	category_name	rental_count	actor_count	total_revenue
103	BUCKET BROTHERHOOD	Travel	34	6	1083.96
738	ROCKETEER MOTHER	Foreign	33	7	816.69
382	GRIT CLOCKWORK	Games	32	3	332.04
730	RIDGEMONT SUBMARINE	New	32	5	653.40
331	FORWARD TEMPLE	Games	32	3	386.04
489	JUGGLER HARDLY	Animation	32	9	870.12
767	SCALAWAG DUCK	Music	32	5	863.40
973	WIFE TURN	Documentary	31	6	1342.14
735	ROBBERS JOON	Children	31	5	528.45
753	RUSH GOODFELLAS	Family	31	5	443.45
621	NETWORK PEAK	Family	31	3	401.07
1000	ZORRO ARK	Comedy	31	3	644.07
891	TIMBERLAND SKY	Classics	31	4	494.76
31	APACHE DIVINE	Family	31	4	714.76
418	HOBBIT ALIEN	Drama	31	8	653.52
369	GOODFELLAS SALUTE	Sci-Fi	31	8	1677.52
239	DOGMA FAMILY	Animation	30	7	1250.90
748	RUGRATS SHAKESPEARE	Action	30	6	424.20
450	IDOLS SNATCHERS	Children	30	8	917.60
374	GRAFFITI LOVE	Sci-Fi	30	6	490.20
609	MUSCLE BRIGHT	Travel	30	5	478.50
563	MASSACRE USUAL	Games	30	3	539.10
341	FROST HEAD	Classics	30	7	536.90
702	PULP BEVERLY	Horror	30	7	956.90
127	CAT CONEHEADS	Comedy	30	4	726.80

film_id	title	category_name	rental_count	total_revenue	revenue_per_rental
892	TITANIC BOONDOCK	Animation	23	1857.24	80.75
680	PINOCCHIO SIMON	Comedy	20	1471.80	73.59
764	SATURDAY LAMBS	Sports	28	1842.48	65.80
827	SPICE SORORITY	Drama	20	1273.80	63.69
938	VELVET TERMINATOR	Comedy	26	1599.66	61.53
879	TELEGRAPH VOYAGE	Music	27	1622.11	60.08
771	SCORPION APOLLO	Drama	23	1350.16	58.70
414	HELLFIGHTERS SIERRA	Foreign	20	1173.60	58.68
694	PREJUDICE OLEANDER	Classics	20	1158.00	57.90
320	FLAMINGOS CONNECTICUT	New	23	1257.93	54.69
551	MAIDEN HOME	New	24	1310.08	54.59
369	GOODFELLAS SALUTE	Sci-Fi	31	1677.52	54.11
112	CALENDAR GUNFIGHT	Foreign	21	1134.32	54.02
846	STING PERSONAL	New	21	1118.53	53.26
263	DURHAM PANKY	Sports	26	1302.66	50.10
172	CONEHEADS SMOOCHY	Drama	25	1230.75	49.23
25	ANGELS LIFE	New	22	1078.02	49.00
637	OPEN AFRICAN	Sci-Fi	21	1024.11	48.77
893	TITANS JERK	Sci-Fi	29	1411.97	48.69
778	SECRETS PARADISE	Family	20	916.80	45.84
10	ALADDIN CALENDAR	Sports	23	1054.16	45.83
579	MINDS TRUMAN	Action	20	898.80	44.94
649	OZ LIAISONS	Animation	24	1053.36	43.89
973	WIFE TURN	Documentary	31	1342.14	43.29
603	MOVIE SHAKESPEARE	Family	27	1165.92	43.18

Query #4: Films performing above the average revenue

Identifies films exceeding the portfolio's average revenue as a quick way to separate stronger than typical performers.

The company should focus retention and recommendation algorithms on above-average revenue films, as they represent high performing assets that contribute disproportionately to overall profitability.

```
SELECT
    film_id, title, category_name,
    total_revenue
FROM dw_film_performance
WHERE total_revenue >
    (SELECT AVG(total_revenue) FROM
    dw_film_performance)
ORDER BY total_revenue DESC;
```

	film_id	title	category_name	total_revenue
▶	892	TITANIC BOONDOCK	Animation	1857.24
	764	SATURDAY LAMBS	Sports	1842.48
	369	GOODFELLAS SALUTE	Sci-Fi	1677.52
	879	TELEGRAPH VOYAGE	Music	1622.11
	938	VELVET TERMINATOR	Comedy	1599.66
	680	PINOCCHIO SIMON	Comedy	1471.80
	893	TITANS JERK	Sci-Fi	1411.97
	771	SCORPION APOLLO	Drama	1350.16
	973	WIFE TURN	Documentary	1342.14
	551	MAIDEN HOME	New	1310.08
	263	DURHAM PANKY	Sports	1302.66
	827	SPICE SORORITY	Drama	1273.80
	320	FLAMINGOS CONNE...	New	1257.93
	239	DOGMA FAMILY	Animation	1250.90
	172	CONEHEADS SMOOCHY	Drama	1230.75
	403	HARRY IDAHO	Drama	1174.20
	414	HELLFIGHTERS SIERRA	Foreign	1173.60
	603	MOVIE SHAKESPEARE	Family	1165.92
	694	PREJUDICE OLEANDER	Classics	1158.00
	112	CALENDAR GUNFIGHT	Foreign	1134.32
	846	STING PERSONAL	New	1118.53
	243	DOORS PRESIDENT	Animation	1114.29
	48	BACKLASH UNDEFEA...	Children	1111.67
	785	SHAWSHANK BUBBLE	Travel	1098.10
	361	GLEAMING JAWBREA...	Sports	1093.60

Query #5: Understocked "Hits" (inventory planning)

Flags popular films with very low inventory levels that may be creating unmet demand; these are strong candidates for adding more copies. Increase inventory for films with high rentals per copy ratios to improve availability and customer satisfaction.

```
SELECT
    film_id, title,
    category_name,
    inventory_count,
    rental_count,
    rentals_per_copy,
    total_revenue
FROM dw_film_performance
WHERE inventory_count <= 8
    AND rental_count >= 30
ORDER BY
    rentals_per_copy DESC,
    total_revenue DESC
LIMIT 25;
```

film_id	title	category_name	inventory_count	rental_count	rentals_per_copy	total_revenue
735	ROBBERS JOON	Children	7	31	4.43	528.45
891	TIMBERLAND SKY	Classics	7	31	4.43	494.76
869	SUSPECTS QUILLS	Action	7	30	4.29	1069.60
450	IDOLS SNATCHERS	Children	7	30	4.29	917.60
979	WITCHES PANIC	Drama	7	30	4.29	694.80
563	MASSACRE USUAL	Games	7	30	4.29	539.10
285	ENGLISH BULWORTH	Sci-Fi	7	30	4.29	518.50
374	GRAFFITI LOVE	Sci-Fi	7	30	4.29	490.20
103	BUCKET BROTHERHOOD	Travel	8	34	4.25	1083.96
738	ROCKETEER MOTHER	Foreign	8	33	4.13	816.69
489	JUGGLER HARDLY	Animation	8	32	4.00	870.12
767	SCALAWAG DUCK	Music	8	32	4.00	863.40
730	RIDGEMONT SUBMARINE	New	8	32	4.00	653.40
331	FORWARD TEMPLE	Games	8	32	4.00	386.04
382	GRIT CLOCKWORK	Games	8	32	4.00	332.04
369	GOODFELLAS SALUTE	Sci-Fi	8	31	3.88	1677.52
973	WIFE TURN	Documentary	8	31	3.88	1342.14
31	APACHE DIVINE	Family	8	31	3.88	714.76
418	HOBBIT ALIEN	Drama	8	31	3.88	653.52
1000	ZORRO ARK	Comedy	8	31	3.88	644.07
753	RUSH GOODFELLAS	Family	8	31	3.88	443.45
621	NETWORK PEAK	Family	8	31	3.88	401.07
239	DOGMA FAMILY	Animation	8	30	3.75	1250.90
403	HARRY IDAHO	Drama	8	30	3.75	1174.20
702	PULP BEVERLY	Horror	8	30	3.75	956.90

Query #6: Category performance summary

Quantifies which categories generate the most revenue and rentals. Based on the information, Sports films generate the most revenue. Allocate more acquisition budget and promotional focus toward the highest performing categories, while reevaluating underperforming categories to determine whether to reduce inventory or refresh content offerings.

```
SELECT
    category_name,
    COUNT(*) AS film_count,
    SUM(total_revenue) AS
category_revenue,
    SUM(rental_count) AS
category_rentals,
    ROUND((SUM(total_revenue) /
NULLIF(SUM(rental_count),0)),2)
AS category_rev_per_rental
FROM dw_film_performance

GROUP BY category_name

ORDER BY category_revenue DESC;
```

category_name	film_count	category_revenue	category_rentals	category_rev_per_rental
Sports	74	32030.55	1179	27.17
Animation	66	26945.80	1166	23.11
Drama	62	25978.48	1060	24.51
Sci-Fi	61	24887.49	1101	22.60
New	63	24149.18	940	25.69
Foreign	73	23947.38	1033	23.18
Action	64	23882.67	1112	21.48
Documentary	68	23580.23	1050	22.46
Comedy	58	21688.33	941	23.05
Family	69	21111.04	1096	19.26
Children	60	20927.90	945	22.15
Horror	56	20261.71	846	23.95
Games	61	20008.58	969	20.65
Travel	57	19836.29	837	23.70
Classics	57	19521.02	939	20.79
Music	51	19002.02	830	22.89

Query #7: High performers that have “gone cold”

Finds historically strong films that have made higher than average revenue but have not been checked out recently, and displays what percentage of customers who have rented the film are still active. This helps determine whether these users can be targeted for advertising to garner more usage of similar titles, or if the lack of recent rental activity instead suggests catalog fatigue, visibility issues, or seasonal demand patterns.

Reintroduce historically strong films through targeted promotions or featured placements to re-engage customers and extend the revenue lifecycle of previously successful titles.

```
SELECT
    film_id, title,
    category_name,
    total_revenue,
    rental_count,
    customer_count,
```

film_id	title	category_name	total_revenue	rental_count	customer_count	active_cust_rate	last_rental_date
431	HOOSIERS BIRDCAGE	Foreign	419.20	8	8	0.88	2005-08-18 00:14:03
43	ATLANTIS CAUSE	Family	789.84	24	24	0.96	2005-08-19 06:38:59
654	PANKY SUBMARINE	Games	886.80	15	15	1.00	2005-08-19 09:34:02
169	COMFORTS RUSH	Travel	424.15	17	17	0.94	2005-08-19 16:50:50
811	SMILE EARRING	Travel	471.20	10	10	1.00	2005-08-19 19:11:41
548	MAGNIFICENT CHITTY	Classics	384.44	8	8	1.00	2005-08-19 19:35:13
679	PILOT HOOSIERS	Family	404.88	16	16	1.00	2005-08-20 05:50:05
448	IDAHO LOVE	New	484.98	17	17	0.82	2005-08-20 08:27:27
678	PICKUP DRIVING	Sports	494.20	16	16	1.00	2005-08-20 12:11:28
626	NOON PAPI	Children	412.98	17	17	0.94	2005-08-20 12:33:36
235	DIVIDE MONSTER	Sci-Fi	406.74	21	21	1.00	2005-08-20 14:23:16
853	STRANGER STRANGERS	Children	683.01	11	11	0.82	2005-08-20 15:18:20
175	CONFUSED CANDLES	Family	616.98	17	17	1.00	2005-08-20 16:17:27
398	HANOVER GALAXY	Music	691.81	17	17	1.00	2005-08-20 18:42:40
661	PAST SUICIDES	Foreign	437.22	13	13	1.00	2005-08-20 21:50:11
521	LIES TREATMENT	Drama	1034.11	27	26	1.00	2005-08-20 23:07:10
396	HANGING DEEP	Drama	796.67	19	19	1.00	2005-08-20 23:42:31
714	RANDOM GO	Sci-Fi	414.83	9	9	1.00	2005-08-20 23:47:54
226	DESTINY SATURDAY	New	677.88	16	16	1.00	2005-08-21 02:56:52
878	TEEN APOLLO	Travel	639.15	17	17	1.00	2005-08-21 03:32:03
84	BOILED DARES	Travel	622.80	15	14	0.93	2005-08-21 04:30:47
473	JACKET FRISCO	Drama	778.20	18	18	1.00	2005-08-21 07:09:27
446	HYSTERICAL GRAIL	Sports	790.65	15	15	1.00	2005-08-21 07:29:05
271	EASY GLADIATOR	Action	452.31	23	22	1.00	2005-08-21 09:06:29
426	HOME PITY	Music	479.16	14	14	1.00	2005-08-21 09:39:50

```

active_cust_rate, last_rental_date
FROM dw_film_performance
WHERE total_revenue >
(SELECT AVG(total_revenue) FROM dw_film_performance)
ORDER BY last_rental_date ASC
LIMIT 25;

```

Query #8 Film performance tiers

This segments the film catalog into performance tiers to support portfolio management. Use portfolio segmentation to guide strategic decisions. Based on this information, we suggest aggressively promoting Blockbusters, maintaining Steady Performers, and evaluating Long Tail films for potential bundling, discounting, or removal to optimize shelf space and capital allocation.

```

SELECT
CASE
    WHEN total_revenue >= 750
THEN 'Blockbuster'
    WHEN total_revenue >= 150
THEN 'Steady Performer'
    ELSE 'Long Tail'
END AS performance_segment,
COUNT(*) AS film_count,
SUM(total_revenue) AS segment_revenue,
ROUND(AVG(rental_count),2) AS avg_rentals
FROM dw_film_performance
GROUP BY performance_segment
ORDER BY segment_revenue DESC;

```

performance_segment	film_count	segment_revenue	avg_rentals
Steady Performer	637	239309.08	17.26
Blockbuster	111	109078.46	22.76
Long Tail	252	19371.13	10.02

3. (50 points) Create **2 useful procedures and 2 useful functions**.

Procedure # 1: Top N films by revenue (returns the top number of films by revenue)

This procedure isolates the highest-earning titles in the catalog to support revenue-driven portfolio optimization. By ranking films based on total revenue and revenue efficiency, it enables management to identify which titles function as core financial engines of the business. These insights guide decisions on inventory allocation, promotional prioritization, and cross-store distribution. Based on this segmentation, the company should aggressively promote top-earning titles, ensure broad availability across all stores, and leverage them in marketing campaigns. Mid-tier performers can be maintained with steady stocking levels, while low-yield titles may be evaluated for bundling, discounting, or selective removal to maximize shelf productivity and capital efficiency.

```

DELIMITER $$

```

```
CREATE PROCEDURE sp_top_films_by_revenue(IN p_limit INT)
```

```
BEGIN
```

```
SELECT
    film_id,
    title,
    category_name,
    rental_count,
    total_revenue,
    ROUND((total_revenue /
    NULLIF(rental_count,0)),2)
AS revenue_per_rental,
    inventory_count,
    rentals_per_copy

FROM dw_film_performance

ORDER BY total_revenue
DESC
```

```
    LIMIT p_limit;
```

```
END $$
```

```
DELIMITER ;
```

```
CALL sp_top_films_by_revenue(25);
```

film_id	title	category_name	rental_count	total_revenue	revenue_per_rental	inventory_count	rentals_per_copy
892	TITANIC BOONDOCK	Animation	23	1857.24	80.75	7	3.29
764	SATURDAY LAMBS	Sports	28	1842.48	65.80	8	3.50
369	GOODFELLAS SALUTE	Sci-Fi	31	1677.52	54.11	8	3.88
879	TELEGRAPH VOYAGE	Music	27	1622.11	60.08	7	3.86
938	VELVET TERMINATOR	Comedy	26	1599.66	61.53	6	4.33
680	PINOCCHIO SIMON	Comedy	20	1471.80	73.59	6	3.33
893	TITANS JERK	Sci-Fi	29	1411.97	48.69	8	3.63
771	SCORPION APOLLO	Drama	23	1350.16	58.70	7	3.29
973	WIFE TURN	Documentary	31	1342.14	43.29	8	3.88
551	MAIDEN HOME	New	24	1310.08	54.59	6	4.00
263	DURHAM PANKY	Sports	26	1302.66	50.10	7	3.71
827	SPICE SORORITY	Drama	20	1273.80	63.69	6	3.33
320	FLAMINGOS CONNE...	New	23	1257.93	54.69	7	3.29
239	DOGMA FAMILY	Animation	30	1250.90	41.70	8	3.75
172	CONEHEADS SMOOCHY	Drama	25	1230.75	49.23	7	3.57
403	HARRY IDAHO	Drama	30	1174.20	39.14	8	3.75
414	HELLFIGHTERS SIERRA	Foreign	20	1173.60	58.68	6	3.33
603	MOVIE SHAKESPEARE	Family	27	1165.92	43.18	6	4.50
694	PREJUDICE OLEANDER	Classics	20	1158.00	57.90	6	3.33
112	CALENDAR GUNFIGHT	Foreign	21	1134.32	54.02	7	3.00
846	STING PERSONAL	New	21	1118.53	53.26	8	2.63
243	DOORS PRESIDENT	Animation	19	1114.29	58.65	6	3.17
48	BACKLASH UNDEFEA...	Children	19	1111.67	58.51	4	4.75
785	SHAWSHANK BUBBLE	Travel	19	1098.10	57.79	6	3.17
361	GLEAMING JAWBREA...	Sports	29	1093.60	37.71	8	3.63

Procedure # 2: Category Dashboard (Returns ranked list of films in your selected category sorted by total revenue and rental count)

This procedure functions as a category-level performance dashboard, enabling management to evaluate the financial strength, rental demand, and inventory efficiency of any selected genre. By combining category-wide totals with a ranked list of individual films, it supports portfolio segmentation within each genre, highlighting which titles act as revenue anchors, which sustain steady engagement, and which underperform relative to peers. These insights guide decisions on genre-specific marketing, targeted promotions, and inventory balancing across stores. High-performing titles within the category should be prioritized for visibility and stocking, while mid-tier films can be maintained at stable levels. Low-yield films may warrant bundling, discounting, or selective retirement to improve category profitability and ensure capital is allocated to the strongest performers.

```
DELIMITER $$
```

```
CREATE PROCEDURE sp_category_film_dashboard(IN p_category VARCHAR(50))
```

```
BEGIN
```

```
-- Category totals
```

```
SELECT
```

```
    category_name,
```

```

COUNT(*) AS film_count,
SUM(total_revenue) AS category_revenue,
SUM(rental_count) AS category_rentals,
(SUM(total_revenue) / NULLIF(SUM(rental_count),0)) AS category_rev_per_rental

FROM dw_film_performance

WHERE category_name = p_category

GROUP BY category_name;


-- Film ranking within that category

SELECT
    film_id,
    title,
    rental_count,
    total_revenue,
    ROUND((total_revenue / NULLIF(rental_count,0)),2) AS revenue_per_rental,
    inventory_count,
    rentals_per_copy,
    last_rental_date

FROM dw_film_performance

WHERE category_name = p_category

ORDER BY total_revenue DESC, rental_count DESC;

END $$

DELIMITER ;

CALL sp_category_film_dashboard('Action');

```

film_id	title	rental_count	total_revenue	revenue_per_rental	inventory_count	rentals_per_copy	last_rental_date
869	SUSPECTS QUILLS	30	1069.60	35.65	7	4.29	2006-02-14 15:16:03
253	DRIFTER COMMANDMENTS	24	992.32	41.35	6	4.00	2006-02-14 15:16:03
838	STAGECOACH ARMAGEDDON	26	928.44	35.71	6	4.33	2005-08-23 16:19:02
162	CLUELESS BUCKET	25	902.00	36.08	7	3.57	2005-08-23 10:25:45
579	MINDS TRUMAN	20	898.80	44.94	7	2.86	2005-08-22 14:27:46
212	DARN FORRESTER	18	844.38	46.91	5	3.60	2005-08-23 21:54:02
21	AMERICAN CIRCUS	22	838.90	38.13	6	3.67	2006-02-14 15:16:03
67	BERETS AGENT	21	787.80	37.51	6	3.50	2006-02-14 15:16:03
964	WATERFRONT DELIVERANCE	17	730.98	43.00	5	3.40	2005-08-21 20:12:30
915	TRUMAN CRAZY	23	730.62	31.77	6	3.83	2005-08-22 21:18:59
911	TRIP NEWTON	28	582.88	20.82	8	3.50	2005-08-23 11:37:32
968	WEREWOLF LOLA	14	552.02	39.43	3	4.67	2005-08-23 20:00:30
292	EXCITEMENT EVE	21	542.32	25.82	6	3.50	2005-08-22 15:52:57
56	BAREFOOT MANCHURIAN	18	534.56	29.70	5	3.60	2005-08-23 19:45:25
542	LUST LOCK	15	523.74	34.92	6	2.50	2006-02-14 15:16:03
271	EASY GLADIATOR	23	452.31	19.67	5	4.60	2005-08-21 09:06:29
395	HANDICAP BOONDOCK	28	446.04	15.93	6	4.67	2005-08-22 18:04:22
501	KISSING DOLLS	20	443.40	22.17	5	4.00	2005-08-23 16:33:33
329	FORREST SONS	18	439.10	24.39	6	3.00	2005-08-23 15:47:02
417	HILLS NEIGHBORS	25	436.50	17.46	6	4.17	2005-08-23 04:17:56
748	RUGRATS SHAKESPEARE	30	424.20	14.14	8	3.75	2005-08-23 20:47:28
229	DEVIL DESIRE	15	419.25	27.95	4	3.75	2006-02-14 15:16:03
574	MIDNIGHT WESTWARD	17	412.47	24.26	5	3.40	2006-02-14 15:16:03
303	FANTASY TROOPERS	26	411.66	15.83	6	4.33	2005-08-22 12:16:46
707	QUEST MUSSOLINI	20	389.00	19.45	6	3.33	2005-08-23 08:19:55

Function # 1: Total revenue for a film by film_ID

This function provides a direct, query-ready revenue KPI for any individual film, enabling analysts and executives to embed financial performance into broader operational workflows. By returning a deterministic revenue value for a selected title, it supports rapid evaluation of film-level profitability, demand forecasting, and contribution analysis across stores and customer segments. This metric becomes especially valuable when integrated into dashboards, recommendation engines, or inventory models that require consistent, reusable financial inputs. With this function in place, management can more easily compare titles, assess whether a film merits additional stocking or promotional investment, and determine which films should be prioritized within the broader portfolio to maximize revenue yield and capital efficiency.

DELIMITER \$\$

CREATE FUNCTION fn_film_total_revenue(p_film_id INT)

RETURNS DECIMAL(10,2)

DETERMINISTIC

READS SQL DATA

BEGIN

DECLARE v_rev DECIMAL(10,2);

SELECT IFNULL(total_revenue,0)

INTO v_rev

revenue_for_film_1
367.70

```

FROM dw_film_performance

WHERE film_id = p_film_id;

RETURN v_rev;

END $$

DELIMITER ;

SELECT fn_film_total_revenue(1) AS revenue_for_film_1;

```

Function # 2: return a category label for a film based on revenue

This function automates portfolio segmentation by assigning each film to a performance tier (Blockbuster, Steady, or Long Tail) based on total revenue. Embedding this logic in the data warehouse ensures consistent classification across dashboards and models. It helps management prioritize high-value titles, support reliable mid-tier films, and manage low-yield assets, strengthening marketing, merchandising, and capital allocation decisions.

```

DELIMITER $$

CREATE FUNCTION fn_film_segment(p_film_id INT)
RETURNS VARCHAR(20)
DETERMINISTIC
READS SQL DATA
BEGIN
    DECLARE v_rev DECIMAL(10,2);
    DECLARE v_seg VARCHAR(20);

    SELECT IFNULL(total_revenue,0)
    INTO v_rev
    FROM dw_film_performance
    WHERE film_id = p_film_id;

    SET v_seg = CASE
        WHEN v_rev >= 750 THEN 'Blockbuster'
        WHEN v_rev >= 150 THEN 'Steady'
        ELSE 'Long Tail'
    END;

    RETURN v_seg;

END $$

```

film_id	title	total_revenue	segment
1	ACADEMY DINOSAUR	367.70	Steady
2	ACE GOLDFINGER	211.72	Steady
3	ADAPTATION HOLES	189.40	Steady
4	AFFAIR PREJUDICE	458.85	Steady
5	AFRICAN EGG	259.40	Steady
6	AGENT TRUMAN	887.53	Blockbuster
7	AIRPLANE SIERRA	414.25	Steady
8	AIRPORT POLLOCK	411.28	Steady
9	ALABAMA DEVIL	646.92	Steady
10	ALADDIN CALENDAR	1054.16	Blockbuster
11	ALAMO VIDEOTAPE	143.04	Long Tail
12	ALASKA PHANTOM	313.18	Steady
13	ALI FOREVER	274.55	Steady
14	ALICE FANTASIA	NULL	Long Tail
15	ALIEN CENTER	544.68	Steady
16	ALLEY EVOLUTION	264.30	Steady
17	ALONE TRIP	502.56	Steady
18	ALTER VICTORY	131.12	Long Tail
19	AMADEUS HOLY	202.74	Steady
20	AMELIE HELLFIGHTERS	407.40	Steady
21	AMERICAN CIRCUS	838.90	Blockbuster
22	AMISTAD MIDSUMMER	283.16	Steady
23	ANACONDA CONFES...	303.95	Steady
24	ANALYZE HOOSIERS	279.30	Steady
25	ANGELS LIFE	1078.02	Blockbuster

```
DELIMITER ;  
  
SELECT film_id, title, total_revenue, fn_film_segment(film_id) AS segment  
FROM dw_film_performance  
ORDER BY title  
LIMIT 25;
```

CHATGPT usage: <https://chatgpt.com/share/69a36df0-b218-8000-8a97-17ac1410f5fd>

ChatGPT was used to help organize the assignment by breaking down the requirements into clear, manageable steps and identifying an efficient workflow for building the data warehouse. It assisted in developing a structured game plan, identifying key performance metrics, and outlining meaningful analytical questions. The tool was used for planning and clarification purposes.